



Higgs Boson at the LHC - Who produced this Higgs?

Master's Degree in Physics

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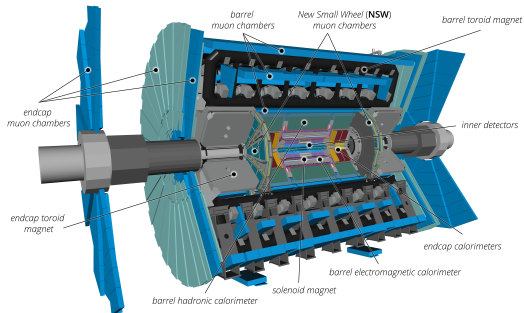
July 21, 2026



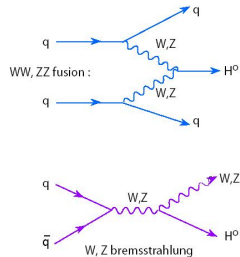
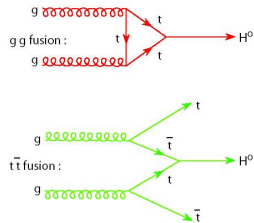
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Higgs Boson Production Mechanisms



Source: atlas.cern



Source: sites.uci.edu



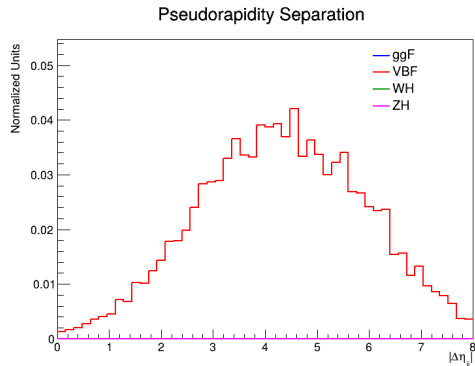
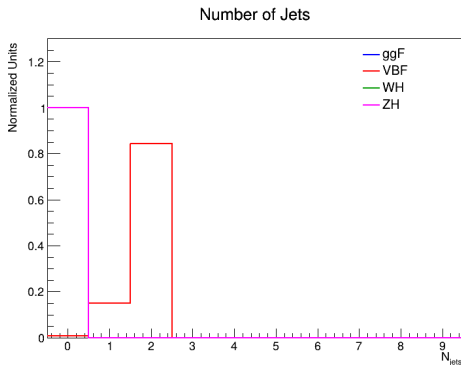
Simulation Workflow: Monte Carlo Generation

Theoretical Setup & Generation Chain:

- **Hard Process:** Generation of the four main Higgs production mechanisms (ggF, VBF, WH, ZH) using MadGraph5_aMC@NLO.
- **Parton-Level (LHE):** Extraction of parton-level kinematics to validate the theoretical predictions and ensure consistency with known characteristics.
- **Shower & Hadronization:** Evolution of hard-scattering events via Pythia8 to simulate parton showering, hadronization, and the production of stable particles.
- **Detector Simulation:** Processing of the hadronized events through Delphes to accurately emulate the ATLAS detector response (efficiencies, smearing, and particle reconstruction).



Simulation Workflow: Monte Carlo Generation





Experimental Signatures at Detector Level

- Following the Delphes simulation, the analysis relies exclusively on reconstructed kinematic variables to mimic experimental conditions.
- **Signal Region (SR) Strategy:** Selection of a single, highly-discriminating variable for each mechanism to optimize the signal-to-background ratio.
- **Golden Variables Identified:**

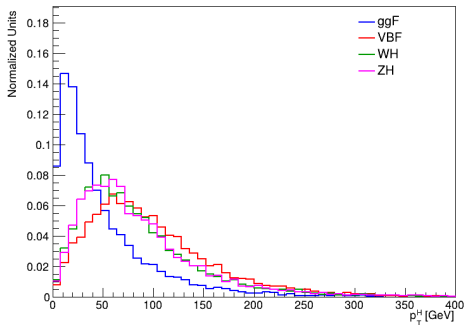
Process	Key Variable	SR Window
ggF	p_T^H	$[0.0, 100.0]$ GeV
VBF	m_{jj}	$[400.0, 1100.0]$ GeV
WH	m_T^W	$[0.0, 120.0]$ GeV
ZH	m_{ll}	$[80.0, 100.0]$ GeV



Experimental Signatures at Detector Level - ggF & VBF

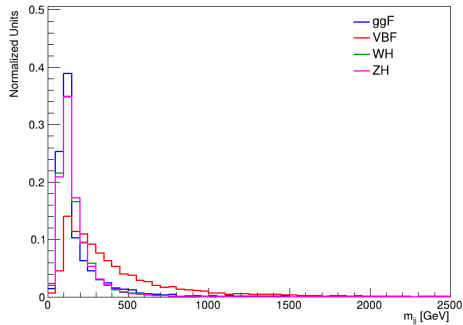
ggF Signature

Higgs Boson p_T



VBF Signature

Dijet Invariant Mass

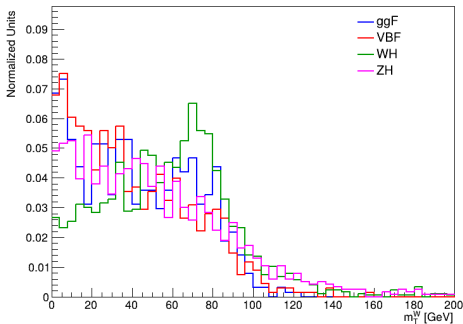




Experimental Signatures at Detector Level - WH & ZH

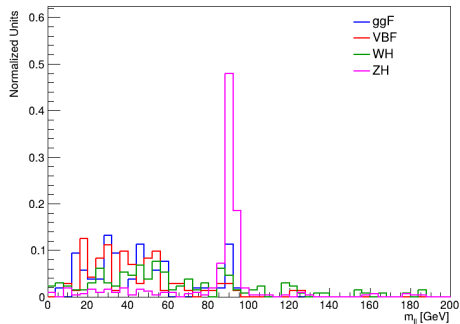
WH Signature

W Transverse Mass



ZH Signature

Dilepton Invariant Mass





Deep Dive: Vector Boson Fusion (VBF) Topology

Background Classification for VBF:

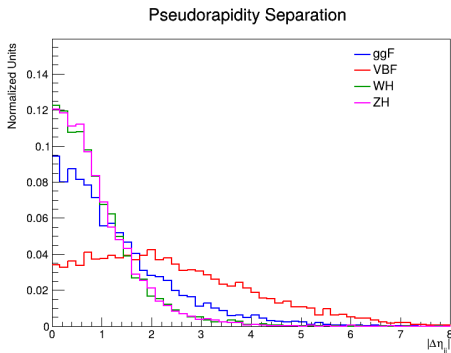
- **Irreducible Background:** $ggF + \geq 2 \text{ jets}$. Same experimental final state.
- **Reducible Backgrounds:** WH / ZH associated production.

Key Separating Observables:

- $\Delta\eta_{jj}$: Large pseudorapidity gap.
- m_{jj} : Hard invariant mass.

Decay-Independent Features:

- The kinematic properties of the forward-backward tagging jets (m_{jj} , $\Delta\eta_{jj}$) depend strictly on the t-channel W/Z exchange of the production vertex.

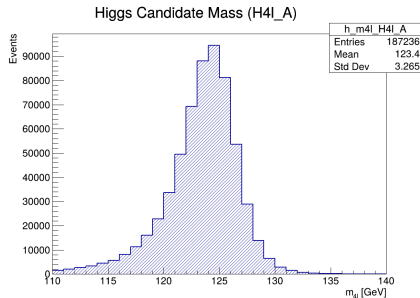




Blind Samples Analysis: Higgs Reconstruction

Analysis Strategy on Blind Samples:

1. **Reconstruction:** Identify the Higgs candidate via its decay products ($H \rightarrow ZZ^* \rightarrow 4\ell$).
2. **Mass Window:** Select events in the Higgs mass peak region ($m_{4\ell} \simeq 125$ GeV).
3. **Kinematics Extraction:** Build distributions for the associated objects (jets, extra leptons, E_T^{miss}).
4. **Template Matching:** Compare with MadGraph MC expectations to deduce the production mechanism.



Invariant mass $m_{4\ell}$ for Sample A clearly showing the Higgs peak.



Signal Regions & VBF Identification

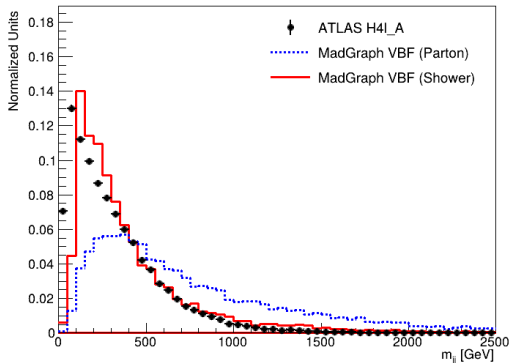
Signal Regions Recall:

- **VBF:** $m_{jj} \in [400, 1100]$ GeV
- **ggF:** $p_T^H \in [0, 100]$ GeV
- **ZH:** $m_{\ell\ell} \in [80, 100]$ GeV
- **WH:** $m_T^W \in [0, 120]$ GeV

Identifying VBF (Sample A):

The dijet invariant mass (m_{jj}) shows excellent shape agreement with the VBF MC template in the high-mass signal region, successfully isolating the signature from irreducible backgrounds

Shape Comparison: m_{jj} (H4I_A vs VBF)



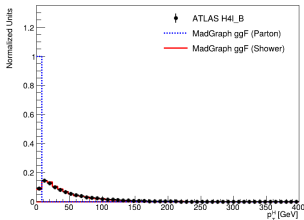


Kinematic Matching for ggF, ZH, and WH

Identifying the remaining blind samples:

ggF (Sample B)

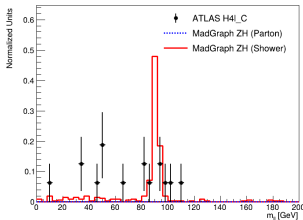
Shape Comparison: $p_{T,H}$ (H4I_B vs ggF)



Key Variable: p_T^H

ZH (Sample C)

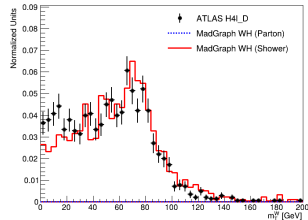
Shape Comparison: $m_{\ell\ell}$ (H4I_C vs ZH)



Key Variable: $m_{\ell\ell}$

WH (Sample D)

Shape Comparison: $m_{T,W}$ (H4I_D vs WH)



Key Variable: m_T^W

Each sample exhibits strong shape compatibility with its respective MC expectation within the specifically defined Signal Region.



Statistical Validation (χ^2 Shape Test)

To quantitatively confirm the visual matches, a χ^2 shape test was performed comparing the Data samples to the MC templates in the corresponding Signal Regions.

Blind Sample	Best MC Match	Key Variable	Shape Test (χ^2/NDF)
Sample A	VBF	m_{jj}	7.08 (Strong Match)
Sample B	ggF	p_T^H	Consistent Shape
Sample C	ZH	$m_{\ell\ell}$	Consistent Shape
Sample D	WH	m_T^W	2.19 (Strong Match)

- The shape tests strongly reject the null hypotheses for mismatched combinations (e.g., Sample D vs ZH yields $\chi^2/\text{NDF} \approx 1000$).
- **Conclusion:** The designed kinematic discriminants successfully and reliably decouple



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Thank you for listening!
Any questions?



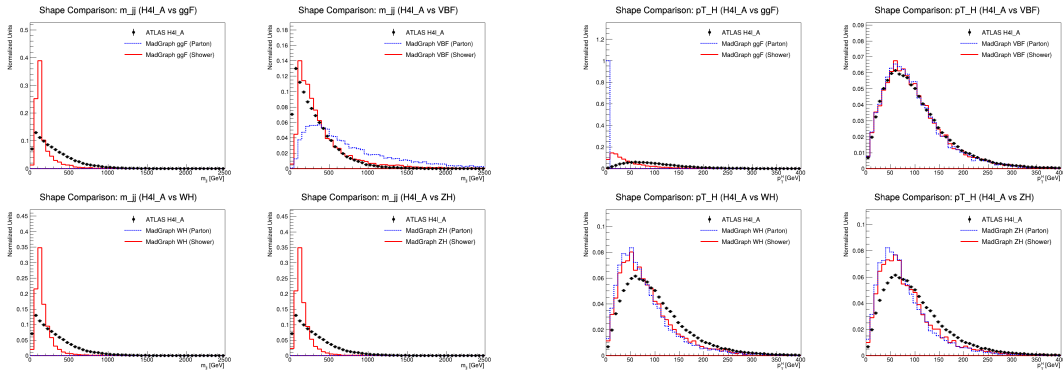
BACKUP



Backup: Extended Kinematic Agreement (VBF)

Sample A vs VBF Expectation

Beyond the key signal region variable (m_{jj}), the full topological profile of the sample is in excellent agreement with the Vector Boson Fusion hypothesis.

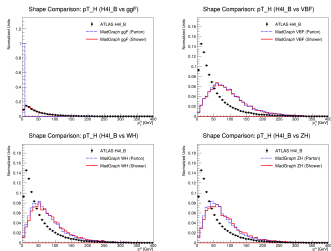




Backup: Hypothesis Rejection for ggF, ZH, WH

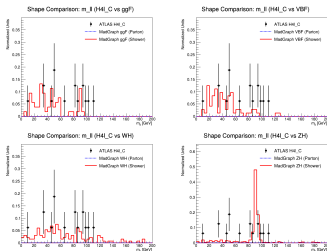
Cross-checking all hypotheses in their respective Signal Regions:

Sample B vs All



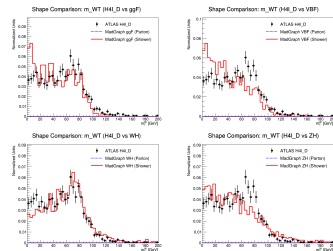
Key: p_T^H . Strong match for **ggF**.

Sample C vs All



Key: $m_{\ell\ell}$. Strong match for **ZH**.

Sample D vs All



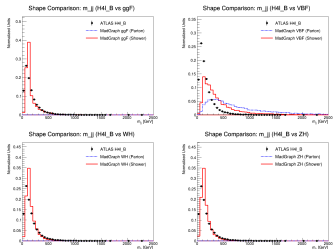
Key: m_T^W . Strong match for **WH**.

The grid comparisons demonstrate that each sample unambiguously prefers a single production mechanism, with alternative hypotheses yielding χ^2/NDF values orders of magnitude larger.



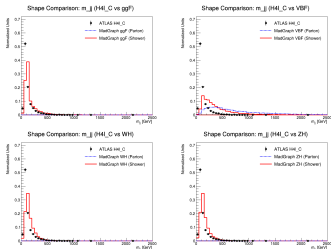
Backup: VBF Signature (m_{jj}) on Non-VBF Samples

Sample B (ggF)



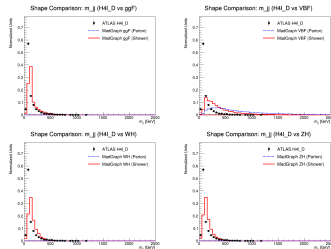
Severe shape mismatch in SR.

Sample C (ZH)



Severe shape mismatch in SR.

Sample D (WH)



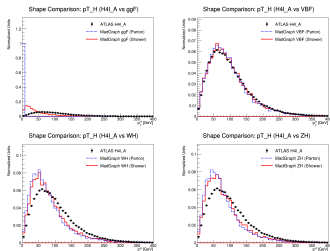
Severe shape mismatch in SR.

Applying the VBF signal region selection ($m_{jj} \in [400, 1100]$ GeV) to non-VBF samples yields drastically poor fits, proving the high rejecting power of the dijet invariant mass against other mechanisms.



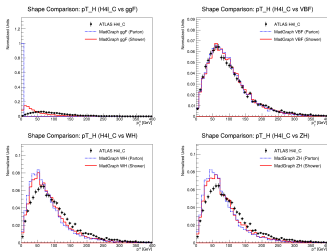
Backup: ggF Signature (p_T^H) on Non-ggF Samples

Sample A (VBF)



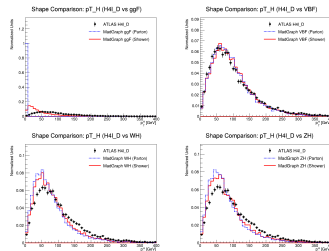
Poor fit across all templates.

Sample C (ZH)



Poor fit across all templates.

Sample D (WH)



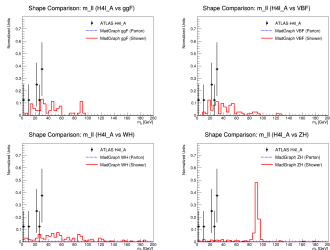
Poor fit across all templates.

The low- p_T^H signal region correctly isolates ggF; when evaluating samples with hard scatterings (VBF, WH) in this regime, the observed kinematics are entirely incompatible with the expected shapes.



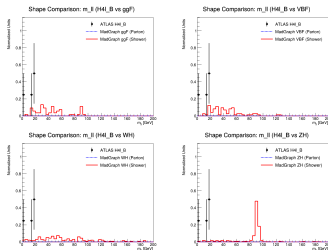
Backup: ZH Signature ($m_{\ell\ell}$) on Non-ZH Samples

Sample A (VBF)



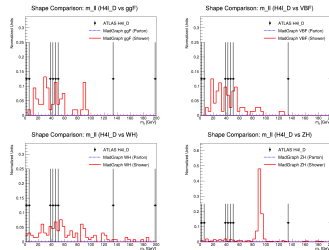
Lack of a clear Z peak.

Sample B (ggF)



Lack of a clear Z peak.

Sample D (WH)



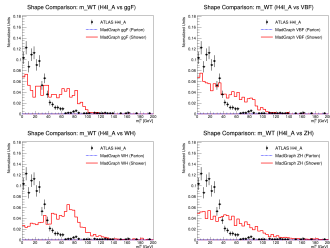
Lacks the extra lepton pair.

The invariant mass of the extra leptons $m_{\ell\ell}$ searches for an associated Z boson. Non-ZH samples fail to populate the expected Z-mass window ([80, 100] GeV), making it an unambiguous discriminant.



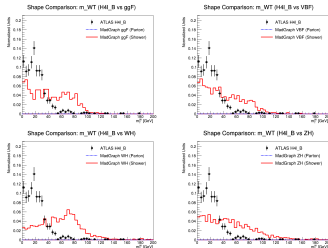
Backup: WH Signature (m_T^W) on Non-WH Samples

Sample A (VBF)



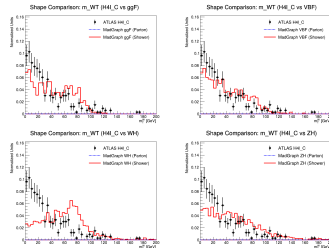
Inconsistent transverse mass.

Sample B (ggF)



Inconsistent transverse mass.

Sample C (ZH)



Inconsistent transverse mass.

The transverse mass m_T^W computed from the extra lepton and E_T^{miss} acts as an excellent veto. Samples lacking an associated W boson production display χ^2 values that completely reject the WH hypothesis.